

Experiment 10:

Write a Python Program to Implement Expectation & Maximization Algorithm

Python Code:

```
import numpy as np
from sklearn.mixture import GaussianMixture
X = np.array([1, 1.2, 1.4, 1.6, 1.8, 5, 5.2, 5.4, 5.6, 5.8]).reshape(-1, 1)
em = GaussianMixture(n_components=2, random_state=42)
em.fit(X)
print("Data Points:\n", X.flatten())
print("\nPredicted Groups:\n", em.predict(X))
print("\nGroup Means:\n", em.means_)
print("\nGroup Probabilities:\n", em.weights_)
```

Output:

```
Data Points:
[1.  1.2 1.4 1.6 1.8 5.  5.2 5.4 5.6 5.8]

Predicted Groups:
[0 0 0 0 0 1 1 1 1 1]

Group Means:
[[1.4]
 [5.4]]

Group Probabilities:
[0.5 0.5]
```